
ENGINEERING DESIGN DOCUMENTATION

LAPTOP STAND

Preliminary Design Specification

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Rev.	Date	Description
rr 0.1	September 2026	Initial preliminary design specification, including design requirements, visual references, preliminary mechanism concept, and material selection criteria and candidate screening.

Design Reference

This section provides visual references for the intended design direction and the existing market context. These images are not engineering drawings and are not intended to define final dimensions or geometry. They serve only as preliminary design references.

Design Inspiration

The following images illustrate the general form, mechanism and aesthetic direction that the proposed laptop stand may take. The final design will be developed independently based on the requirements and engineering constraints established in this document.

Existing Market Product

The following image shows an existing commercially available laptop stand used as a market reference. Its geometry, mechanism and functionality may be studied for comparison, but the final design will be developed independently.



Figure 1: Reference concept illustrating the intended design direction.



Figure 2: Existing commercial laptop stand used as a preliminary market and mechanism reference.

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1 Purpose

The purpose of this project is to design and manufacture a **lightweight, portable, compact and mechanically robust laptop stand** capable of supporting laptops in the approximate mass range of **2–3 kg**.

The design shall provide adjustable height and width while maintaining sufficient structural stiffness and stability during normal use. The stand shall also be designed with manufacturability and portability as primary considerations.

A preliminary factor of safety (FoS) of

$$\text{FoS} = 1.5\text{--}2.0$$

shall be used during the initial structural design and analysis stages. The final factor of safety will be established after the material, geometry, loading conditions and manufacturing method have been finalised.

The project is intended not only to produce a functional physical product, but also to provide a complete engineering design workflow covering:

- requirements definition,
- mechanical concept development,
- material selection,
- CAD modelling,
- mechanism design,
- structural simulation,
- fatigue assessment,
- manufacturing considerations,
- engineering drawings,
- physical fabrication and testing,
- and version-controlled engineering documentation.

2 Functional Requirements

The laptop stand shall satisfy the following primary functional requirements.

1. Support a laptop with a nominal mass of approximately **2–3 kg** without structural failure.
2. Maintain sufficient stiffness such that deflection under normal loading does not adversely affect laptop stability or user operation.
3. Provide **8 discrete height adjustment modes**.
4. Prevent or minimise sliding of the laptop during normal use.

- 5. Remain stable on a supporting surface during loading and adjustment.
- 6. Collapse into a substantially smaller configuration for storage and transportation.
- 7. Permit adjustment of the stand without requiring excessive force or complex tools.
- 8. Use a mechanical architecture that can reasonably be manufactured using accessible manufacturing processes.
- 9. Maintain functionality through repeated adjustment cycles.

2.1 Functional Requirement Summary

rr	ID	Category	Requirement	Status
	FR-01	Load capacity	Support laptops in the nominal 2–3 kg mass range.	Open
	FR-02	Safety	Meet a preliminary factor of safety of 1.5–2.0.	Open
	FR-03	Height adjustment	Provide 8 discrete height modes.	Defined
	FR-04	Slip prevention	Prevent unacceptable laptop movement during operation.	Open
	FR-05	Collapsibility	Allow the stand to fold into a compact configuration.	Defined
	FR-06	Structural stiffness	Limit deflection to an acceptable level under design loading.	Open
	FR-07	Durability	Maintain function over repeated adjustment cycles.	Open
	FR-08	Manufacturability	Use practical and accessible manufacturing processes.	Open

3 Dimensional Requirements

The following dimensions define the initial design envelope.

Table 1: Preliminary dimensional requirements

Parameter	Preliminary Range	Notes
rr Height, h	2–10 cm	Adjustable operating height.
Length, l	5–25 cm	Preliminary adjustable dimension.
Width, w	25–30 cm	Final value TBD based on laptop compatibility and mechanism geometry.

All dimensional values listed above are **preliminary estimates**. They shall not be treated as final design constraints until the mechanism, laptop envelope, structural requirements and

manufacturing process have been evaluated.

Dimensional values may therefore change during subsequent iterations.

3.1 Dimensional Design Considerations

The final dimensions shall be determined based on:

- target laptop size range,
- required stability,
- height adjustment mechanism,
- width adjustment mechanism,
- hinge and pin dimensions,
- material thickness,
- manufacturability,
- portability,
- structural stiffness,
- and interference between moving components.

4 Load Requirements

The primary design load shall represent the weight of a laptop acting on the supporting surfaces of the stand.

The preliminary laptop mass range is

$$m = 2\text{--}3 \text{ kg}.$$

The corresponding gravitational load is

$$F = mg,$$

where

$$g = 9.81 \text{ m/s}^2.$$

Therefore,

$$F_{2\text{ kg}} \approx 19.62 \text{ N}$$

and

$$F_{3\text{ kg}} \approx 29.43 \text{ N}.$$

The nominal maximum static laptop load is therefore approximately

$$F_{\text{nominal,max}} \approx 29.4 \text{ N}.$$

The structural design shall subsequently account for the selected factor of safety.

The load is expected to act primarily through approximately **six contact points** between the laptop and the supporting structure.

4.1 Load Distribution

For initial analysis, the six contact points may be represented as discrete or distributed reaction locations depending on the geometry of the final support structure.

The actual load distribution shall be investigated during CAD and FEA development rather than assumed to be perfectly uniform.

Potential non-uniformity may result from:

- laptop mass distribution,
- centre-of-gravity location,
- support geometry,
- component deformation,
- manufacturing tolerances,
- and user-applied loads.

4.2 Design Loading Cases

The structural analysis shall eventually consider, at minimum:

1. nominal static laptop loading,
2. maximum nominal laptop loading,
3. loading combined with the selected factor of safety,
4. asymmetric loading,
5. loading at critical adjustment positions,
6. and repeated loading for fatigue assessment.

5 Adjustment and Collapsibility Requirements

The stand shall be capable of changing its operating configuration while remaining mechanically stable.

The primary adjustment requirements are:

- adjustable height,
- adjustable width,

- controlled discrete height positions,
- collapsibility for storage,
- and repeatable positioning.

The preliminary design shall provide

8 discrete height positions

while allowing the structure to collapse when not in use.

Height and width adjustment are initially intended to be achieved using hinged and sliding mechanical elements.

The final adjustment mechanism shall be evaluated for:

- range of motion,
- locking reliability,
- ease of adjustment,
- structural stiffness,
- wear,
- manufacturing complexity,
- and accidental disengagement.

6 Material Requirements

The stand shall primarily use a metallic material or combination of metallic materials capable of sustaining the expected loads with acceptable deformation and fatigue performance.

The material selection shall consider:

- yield strength,
- ultimate tensile strength,
- elastic modulus,
- density,
- fatigue strength,
- corrosion resistance,
- manufacturability,
- availability,
- cost,
- surface finish,

- and suitability for the selected manufacturing processes.

The material must be capable of carrying the design load for extended periods while also tolerating repeated loading and adjustment cycles without unacceptable permanent deformation or fatigue failure.

6.1 Material Selection Method

Material candidates shall be identified using an appropriate engineering material database.

The final selection shall be justified using a combination of:

1. mechanical performance,
2. mass,
3. manufacturing requirements,
4. cost,
5. availability,
6. and simulation results.

The selected material shall be recorded in the final design documentation together with its relevant mechanical properties and source.

7 Material Selection and Candidate Screening

Material selection was treated as a design problem rather than simply selecting the material with the highest strength. The stand is required to support a 2–3 kg laptop while remaining lightweight, sufficiently stiff, manufacturable and economical. In addition, the mechanism contains hinges, pins and sliding interfaces that will experience repeated loading and adjustment cycles.

The material selection process therefore considered both the bulk structural requirements and the requirements imposed by the individual mechanical interfaces.

7.1 Selection Criteria

The principal criteria used to screen candidate materials were:

1. **Yield strength:** The material must have sufficient strength to prevent permanent deformation under the design load and selected factor of safety.
2. **Elastic modulus:** Since excessive deformation is one of the primary design concerns, stiffness is an important selection criterion in addition to strength.
3. **Density:** The stand should remain lightweight and portable. Materials with unnecessarily high density therefore incur a mass penalty.
4. **Fatigue performance:** The stand will experience repeated loading and repeated adjustment cycles. The material must therefore be suitable for cyclic loading.
5. **Corrosion resistance:** Since the stand is intended for long-term use, resistance to environmental degradation is desirable.

6. **Manufacturability:** The material should be compatible with the anticipated manufacturing processes, particularly sheet bending, machining and drilling.
7. **Cost:** Material cost is an important design constraint. A material that provides substantially more performance than required may not be justified if a cheaper alternative satisfies the engineering requirements.
8. **Availability:** The material should be reasonably accessible for prototype fabrication and future replacement parts.

7.2 Initial Engineering Screening

The material screening was performed after considering the preliminary load path and the expected loading of the critical mechanical components.

The maximum nominal laptop load is approximately

$$F = mg = 3(9.81) \approx 29.4 \text{ N}.$$

This load is relatively small compared with the capacity of most engineering metals. Consequently, bulk tensile strength alone is not expected to be the dominant design constraint.

Instead, particular attention was given to the behaviour of the critical joints and load-bearing features, including:

- pin shear and bending,
- bearing stress at joints,
- local deformation,
- bending of slender components,
- stress concentration around adjustment holes and slots,
- sliding-joint loading,
- and overall structural deflection.

A preliminary yield-strength screening value was also considered during the material database search. This value was used as an initial screening criterion rather than as a final pass/fail requirement, since the final suitability of a material depends on the complete geometry, loading condition and manufacturing process.

The screening therefore aimed to identify a range of materials rather than immediately selecting a single material.

7.3 Selection Strategy

The final candidate set was intentionally constructed to represent different points on the engineering trade-off space.

Rather than selecting only the strongest available materials, the candidate set includes:

1. a balanced and widely applicable baseline material,

- 2. higher-performance materials that provide additional strength or corrosion resistance at increased cost,
- 3. and materials that are close to or below the preliminary screening criteria.

This approach allows the effect of material selection to be evaluated quantitatively during the subsequent finite element analysis.

In particular, all candidate materials can later be evaluated using the same CAD geometry, boundary conditions and loading conditions. Stress, deformation, factor of safety, mass and other relevant performance parameters can then be compared directly.

7.4 Final Material Candidates

The material screening resulted in the following five candidates for further engineering evaluation.

Table 2: Final material candidates selected for further analysis

Material	Selection Role	Reason for Inclusion
6061-T6 Aluminium	Baseline	Balanced combination of strength, stiffness, low density, manufacturability, availability and cost. Serves as the primary reference material for the design.
7075-T6 Aluminium	High-performance	Higher-strength aluminium candidate retained to determine whether the additional mechanical performance provides a meaningful benefit over 6061-T6.
rr 316 Stainless Steel	High-performance	High-strength, corrosion-resistant candidate included to evaluate the trade-off between structural performance, durability, density and cost.
5052-H32 Aluminium	Near-miss	Lower-strength aluminium candidate retained to determine whether a more easily formed and potentially lower-cost alloy can still satisfy the structural requirements.
304L Stainless Steel	Near-miss	Stainless-steel comparison candidate with useful corrosion resistance and manufacturability, retained despite being less favourable against the preliminary strength screening criterion.

7.5 Purpose of the Five-Material Comparison

The five candidates are not intended to represent five equally likely final materials. Instead, they provide a useful engineering comparison.

The expected comparison can be represented conceptually as:

6061-T6 ↔ 7075-T6 / 316 ↔ 5052-H32 / 304L

where 6061-T6 represents the balanced baseline, 7075-T6 and 316 represent higher-performance

alternatives, and 5052-H32 and 304L provide lower-performance comparison cases.

This allows the subsequent analysis to determine whether the additional strength, corrosion resistance or other properties of the premium materials provide a meaningful engineering advantage.

Conversely, the lower-performing candidates allow the analysis to demonstrate whether the preliminary geometry has sufficient structural margin even when a less favourable material is used.

7.6 Material Selection and FEA

The final material shall not be selected solely from the database screening.

The five candidates will subsequently be evaluated using finite element analysis under equivalent loading and boundary conditions. The comparison shall consider, where applicable:

- maximum von Mises stress,
- maximum deformation,
- factor of safety,
- component mass,
- critical joint stresses,
- fatigue performance,
- corrosion considerations,
- manufacturing suitability,
- and relative material cost.

The final material decision will therefore be based on the combination of material properties, structural simulation and practical manufacturing considerations.

Final material selection: to be determined following CAD and finite element analysis.

8 Manufacturing Requirements

The design shall favour manufacturing processes that are accessible, repeatable and appropriate for relatively small-scale fabrication.

The initial manufacturing concept is based primarily on:

- sheet-metal bending,
- milling,
- drilling,
- cutting,
- and conventional fastening and joining methods.

However, the manufacturing process shall remain an open design constraint until the component geometries and material selection have been established.

The design shall therefore avoid unnecessary geometric complexity where possible.

8.1 Manufacturability Considerations

During CAD development, the following shall be considered:

- minimum bend radii,
- sheet thickness,
- machining accessibility,
- hole tolerances,
- pin and hinge fits,
- assembly sequence,
- component accessibility,
- fastening method,
- manufacturing tolerances,
- and material availability.

The final manufacturing plan shall be developed after the component architecture has stabilised.

Final manufacturing process: TBD.

9 Component Breakdown

The stand shall be decomposed into individual components such that each component has a clearly defined mechanical function, interface and manufacturing requirement.

The detailed component breakdown shall be maintained as the mechanical architecture develops.

The current conceptual architecture includes the following functional component groups:

1. Laptop support structure
2. Base structure
3. Height-adjustment arms
4. Cross-joint arms
5. Hinge assemblies
6. Sliding width-adjustment elements
7. Pins, axles and fasteners
8. Anti-slip contact elements

The detailed component list, individual part numbers, material assignments and manufacturing methods shall be developed in the subsequent CAD design stage.

9.1 Component Documentation

Each final component should eventually have:

- unique part identifier,
- part name,
- material,
- manufacturing process,
- primary dimensions,
- relevant tolerances,
- interfaces,
- loading conditions,
- and associated engineering drawing.

Detailed component breakdown: to be developed from the mechanical concept.

10 Mechanical Interfaces

The design contains several important mechanical interfaces. These interfaces shall be explicitly defined because they determine the load path and constrain the motion of the mechanism.

Table 3: Preliminary mechanical interfaces

Interface		Function
rr	Laptop → Stand	Transfers laptop loading to the supporting structure while preventing unacceptable sliding.
	Stand → Base	Transfers structural loads and provides the primary mechanical connection between the upper stand and base.
	Base → Floor	Transfers the resulting loads to the supporting surface and provides overall stability.
	Support Arm → Base	Provides a constrained pivoting connection and transfers load into the base structure.
	Support Arm → Stand	Provides the mechanical connection required for height adjustment and structural support.

The final design shall ensure that all interfaces have sufficient strength and stiffness and that the expected range of motion is not restricted by unintended interference.

11 Initial Mechanism Concept

The initial mechanism is based on a combination of rotational and translational joints.

The preliminary joint architecture is:

The mechanism is therefore initially composed of

2 base hinges + 2 height hinges + 1 cross pin + 2 sliding joints

Table 4: Initial joint architecture

Joint Type	Quantity	Purpose
Base hinge joints	2	Permit folding and rotation of the primary structure relative to the base.
^{rr} Height-arm hinge joints	2	Provide rotational movement for height adjustment.
Cross-joint pin	1	Connects the cross-joint arms and constrains their relative motion.
Width sliding joints	2	Permit lateral adjustment of the stand width.

This represents the initial kinematic concept rather than the final mechanism.

The joint locations, degrees of freedom, travel limits, locking mechanism and physical dimensions shall be established during CAD development.

11.1 Mechanism Design Objectives

The mechanism shall:

- provide the required eight height positions,
- permit controlled collapse,
- permit width adjustment,
- resist unwanted motion during loading,
- avoid mechanical interference,
- minimise backlash,
- remain sufficiently stiff,
- and be practical to manufacture and assemble.

12 Simulation and Analysis Objectives

Engineering simulation shall be used to validate the mechanical design before physical fabrication.

The primary simulation objectives are:

12.1 11.1 Range of Motion Analysis

The mechanism shall be evaluated across its complete intended range of motion.

The analysis shall verify:

- all eight height positions,
- width adjustment range,
- collapsed configuration,

- maximum and minimum geometric positions,
- joint travel,
- and absence of unintended interference.

12.2 11.2 Static Structural Analysis

Static structural analysis shall be performed to determine:

- maximum stress,
- maximum deformation,
- reaction forces,
- critical components,
- critical joints,
- and resulting factor of safety.

The critical loading configuration shall be determined from the geometry rather than assuming that the nominal operating position is necessarily the worst case.

12.3 11.3 Fatigue Analysis

Fatigue analysis shall evaluate the effect of repeated loading and adjustment cycles.

The analysis shall consider:

- cyclic laptop loading,
- repeated adjustment,
- stress concentrations,
- hinge and pin loading,
- and the expected service life of critical components.

The final fatigue requirement shall be established once an expected usage cycle count has been determined.

Required service cycle count: TBD.

12.4 11.4 Simulation Software

The exact simulation workflow is expected to involve the engineering software used for CAD, mechanism analysis and finite element analysis.

The final software stack shall be documented as the project develops.

13 Design Constraints and Open Parameters

The following parameters are intentionally left open at this stage.

Table 5: Open design parameters

Parameter	Current Status
Final stand width	TBD
Final height positions	Eight required; exact dimensional positions TBD.
Final length adjustment	Preliminary range 5–25 cm; TBD.
Material	TBD through material-selection study.
rr Sheet thickness	TBD.
Pin and hinge diameter	TBD through structural and manufacturing analysis.
Joint clearances	TBD.
Anti-slip material	TBD.
Manufacturing process	Preliminary sheet bending and milling; TBD.
Fatigue service life	TBD.
Final factor of safety	Preliminary range 1.5–2.0.

These parameters shall be updated as the design progresses. Changes shall be tracked through the project's version-control system.

14 Version-Control Strategy

Git and GitHub shall be used as the primary version-control system for the engineering project. The objective is to maintain a traceable history of design development across all major engineering disciplines rather than only source code.

Version control is expected to cover:

- CAD models,
- assemblies,
- engineering drawings,
- LaTeX documentation,
- MATLAB scripts and models,
- ANSYS project files where practical,
- simulation data where appropriate,
- design calculations,
- supporting engineering files,
- and other project documentation.

14.1 CAD Version Control

SolidWorks shall be the primary CAD platform.

Engineering drawings generated through SolidWorks shall be maintained under the same project version-control strategy.

Large binary files and generated files shall be handled according to the project’s Git storage strategy to prevent unnecessary repository growth.

14.2 Documentation Version Control

This LaTeX document shall form the central design documentation record.

Design decisions, requirement changes, material selections and major geometry revisions shall be reflected in the documentation as the project progresses.

14.3 Revision Philosophy

Major design changes shall receive identifiable revisions or Git commits.

The objective is to preserve the reasoning and evolution of the design, not merely the final geometry.

15 Preliminary Design Requirement Matrix

The following matrix provides a consolidated view of the current requirements.

Table 6: Preliminary design requirement matrix

ID	Parameter	Requirement	Status
FR-01	Load capacity	Support a 2–3 kg laptop.	Required
FR-02	Factor of safety	Preliminary factor of safety of 1.5–2.0.	Preliminary
FR-03	Height adjustment	Eight discrete height modes.	Required
FR-04	Height range	Approximately 2–10 cm.	Preliminary
FR-05	Length range	Approximately 5–25 cm.	Preliminary
FR-06	Width	Approximately 25–30 cm.	Preliminary
FR-07	Slip prevention	Prevent unacceptable laptop movement.	Required
FR-08	Collapsibility	Fold into a compact configuration.	Required
FR-09	Width adjustment	Adjustable through a sliding mechanism.	Required
FR-10	Structural stiffness	Low deformation under design loading.	Required
FR-11	Durability	Survive repeated loading and adjustment.	Required

ID	Parameter	Requirement	Status
FR-12	Material	Metallic material with suitable strength, stiffness, fatigue performance and manufacturability.	TBD
FR-13	Manufacturing	Prefer sheet bending and milling where practical.	Preliminary
FR-14	Static analysis	Evaluate stress, deformation and factor of safety.	Required
FR-15	Fatigue analysis	Evaluate repeated loading and service life.	Required
FR-16	Motion analysis	Verify full range of mechanism motion.	Required
FR-17	Documentation	Maintain engineering documentation in LaTeX.	Required
FR-18	Version control	Use Git and GitHub for engineering project files.	Required

16 Preliminary Design Philosophy

The design shall follow an iterative engineering process rather than attempting to establish all final dimensions at the beginning of the project.

The intended design progression is:

Requirements → Concept → Mechanism → Material → CAD → Simulation → Manufacturing → Testing

Each stage may result in modifications to earlier decisions.

In particular, dimensional requirements, material selection, joint geometry and manufacturing methods shall be treated as coupled design variables.

A design change that improves one property may therefore require re-evaluation of other properties.

For example:

Reducing mass → lower section thickness → increased deflection → geometry modification

Similarly:

Increasing hinge compactness → smaller pin diameter → higher bearing stress → material / geometry revision

This iterative approach is intentional and forms a central part of the engineering design process for the project.

17 Next Design Activities

The immediate design activities following this preliminary specification are:

1. Finalise the functional component breakdown.
2. Establish the preliminary load path.
3. Define the degrees of freedom of the mechanism.
4. Establish the required eight height positions.
5. Determine the preliminary width-adjustment mechanism.
6. Identify candidate materials.
7. Compare candidate materials using engineering material data.
8. Develop the first CAD concept.
9. Define preliminary sheet thicknesses and pin dimensions.
10. Perform an initial range-of-motion study.
11. Identify critical structural components.
12. Perform preliminary static structural analysis.
13. Iterate the geometry based on simulation results.
14. Establish the manufacturing process.
15. Produce engineering drawings.
16. Fabricate and physically test the prototype.

The design specification shall be updated as major design decisions are made.

18 Conclusion

This document establishes the preliminary engineering requirements for the laptop stand project.

The initial concept is a compact, adjustable and collapsible metallic structure capable of supporting a 2–3 kg laptop with a preliminary factor of safety between 1.5 and 2.0.

The design incorporates:

- eight discrete height adjustment modes,
- adjustable width,
- a collapsible structure,
- anti-slip contact points,

- hinged support members,
- a cross-joint mechanism,
- sliding width-adjustment elements,
- and a mechanically stable load path.

At this stage, dimensions, materials, manufacturing methods and component details remain subject to engineering evaluation.

The specification therefore serves as a **baseline** from which the CAD, material-selection, simulation, manufacturing and testing stages can proceed.

All subsequent changes shall be documented and version-controlled so that the final design retains a traceable engineering development history.

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